

## Lesson 15: Direct Comparison

### Assessment Criteria

- **Positive Terms Verification** (5%) : Confirm that both series  $\sum a_n$  and  $\sum b_n$  have positive terms ( $a_n, b_n \geq 0$ ) for  $n > N$ .
- **Valid Inequality Establishment** (5%):
  - **For Convergence:** Demonstrate  $0 \leq a_n \leq b_n$  (the test series is smaller than a known convergent series).
  - **For Divergence:** Demonstrate  $0 \leq b_n \leq a_n$  (the test series is larger than a known divergent series).
- **Correct Comparison Selection** (50%): Select a known series  $\sum b_n$  (often a geometric series  $|r| < 1$  or  $p$ -series  $p > 1$ ) that has the same end behavior as the dominant terms of  $a_n$ .
- **Justification of Inequality** (20%): Provide algebraic justification (e.g.,  $n^2 + 1 > n^2$  or  $\sin^2(n) \leq 1$ ) to prove the inequality holds for all  $n > N$ .
- **Conclusion Formulation** (20%): Clearly state whether the series converges or diverges based on the comparison.